MIL-F-10969B(MU)
17 December 1969
SUPERSEDING
MIL-F-10969A(CmlC)
2 November 1955

MILITARY SPECIFICATION

FILTER ELEMENT, AEROSOL

1. SCOPE

1.1 This specification covers one type of protective mask canister aerosol filter element.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-530	-	Plywood, Flat-Panel.
QQ-S-781	-	Steel, Strapping, Flat.
TT-W-00572	-	Wood Preservative, Water-Repellent.
PPP-B-601	-	Boxes; Wood, Cleated-Plywood.
PPP-B-621	-	Boxes; Wood, Nailed and Lock Corners.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-B-2427	-	Boxes, Ammunition Packing: Wood, Nailed.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL5-3-588	-	Canister, M11
DLB136-42-102-5	-	DOP Filter Testing Penetrometer, Q127.
D150-2-1	-	Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF COMMERCE

CS-35 - Hardwood Plywood.

(Application for copies should be addressed to the US Government Printing Office, Washington, D. C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N.W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Materials. All materials for the filter element, cited on Drawing DL5-3-588 and subsidiary drawings, shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

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3.2 Manufacture and assembly. The filter element (see 6.3) shall be manufactured and assembled in accordance with Drawing C5-3-583 and subsidiary drawings listed thereon.

3.3 Adhesion. The filter material shall be thoroughly and completely bonded with the paper tab at all points of contact and with itself at the center pleat. All cemented joints shall be essentially as strong as the filter material itself when tested as specified in 4.4.4.1. The filter material shall tear within itself rather than pulling loose and the filter material fibers shall remain securely attached to the adhesive.

3.4 Airflow resistance. The airflow resistance of the filter element shall not exceed 10 millimeters of water at a flow-rate of 32 liters per minute when tested as specified in 4.4.4.2.

3.5 DOP smoke penetration. DOP smoke penetration of the filter element shall not exceed 0.015 percent at a flow-rate of 32 liters per minute when tested as specified in 4.4.4.3.

3.6 Preproduction. Prior to the start of regular production a preproduction sample of filter elements shall be produced in accordance with this specification for examination and tests (see 4.3).

3.7 Workmanship. The filter element shall be free from foreign matter, contamination, or damage such as tears, punctures, or that which might affect its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has been provided in this specification have been satisfied.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 500 filter elements shall be produced using the same methods, materials, and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. Each filter element of the preproduction sample shall be examined and tested in accordance with the classification of defects (4.4.3.3).

4.3.2.2 For destructive tests. Forty-four filter elements, taken at random from the preproduction sample, shall be tested as specified in 4.4.4.1.

4.3.3 Acceptance/rejection criteria. If any filter element in the preproduction sample fails to meet all the requirements of this specification then the preproduction sample shall be rejected. The supplier shall obtain written approval from the contracting office before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the filter elements produced by one manufacturer, from the same type of filter material, having the same number of pleats in each element, and under essentially the same manufacturing conditions. No more than one lot of filter material shall be represented in any one lot of finished filter elements.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Sixty filter elements, 20 for test and 40 for retest, shall be taken at random from each lot.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample filter elements shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

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4.4.3.2 For destructive tests. Twenty filter elements from the sample of 60 shall be tested for adhesion in accordance with 4.4.4.1. If the test fails then the remaining 40 shall also be tested (see 4.4.5).

4.4.3.3 Classification of defects.

(a) Aerosol filter element (Dwg. C5-3-583 and C5-3-584, C5-3-668, or C5-3-670).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.25 percent defective	
101	Airflow resistance	4.4.4.2
102	DOP smoke penetration AQL 1.0 percent defective	4.4.4.3
103	Component missing or incorrectly assembled	
104	Overall diameter incorrect	
105	Overall height incorrect	
106	Workmanship (damage) (3.7)	
Minor:	AQL 2.5 percent defective	
201	Height outside fold incorrect	
202	Workmanship foreign matter or contamination) (3.7)	

4.4.4 Tests.

4.4.4.1 Adhesion. Adhesion of the element shall be determined by applying tension at all the cemented joints and tearing the assembly apart.

4.4.4.2 Airflow resistance. Airflow resistance of the filter element shall be determined with the Q127 DOP Filter Testing Penetrometer (DLB 136-42-102-5).

4.4.4.3 DOP smoke penetration. DOP smoke penetration of the filter element shall be determined with the Q127 DOP Filter Testing Penetrometer (DLB 136-42-102-5).

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4.4.5 Acceptance/rejection criteria. If the first 20 sample elements pass the adhesion test, the lot represented is acceptable. If three or more of the 20 elements fail, the lot shall be rejected. If one or two of the 20 elements fail, then a retest shall be conducted on the remaining 40 sample elements. If three or more elements fail adhesion in the test and retest combined, then the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. A filter ring, conforming to Drawing C5-3-558 shall be inserted in the filter element inside the outermost fold. The filter and ring assembly shall then be wrapped in tissue or glassine paper.

5.1.1 Intermediate packaging. Five hundred filter and ring assemblies packaged as specified in 5.1 shall be packaged in a style RSC fiberboard box conforming to W5c or W6c of PPP-B-636. The filter element shall be arranged in 20 layers, each containing 5 rows of 5 filter elements each. The filter elements shall be separated from one another by fiberboard interlocking partitions and pads between layers fabricated from material conforming to type CF, class weather-resistant, of PPP-F-320. The box shall be closed and sealed in accordance with the appendix thereto.

5.2 Packing, level A. Two intermediate packages, packaged as specified in 5.1.1 shall be packed in a style 4 nailed wood box conforming to PPP-B-621, or in an overseas type, style A, pleated plywood box conforming to PPP-B-601, and using plywood complying to group A of NN-P-530 in accordance with type I, grade 3 or 4 of CS-35, provided the plywood does not have continuous holes or openings extending the entire panel from face to back. The box shall be closed and strapped in accordance with the appendix thereto. Strapping conforming to type I, class B of QQ-S-781 shall be used. The boxes or the finished wood parts thereof, shall be completely immersed for a minimum of three minutes in a water repellent wood preservative conforming to type II, composition A of TT-W-00-572. Prior to packing the items in the box, the treated wooden box shall meet the preservative dryness requirement of MIL-B-2427.

5.2.1 Packing, level B. Two intermediate packages, packaged as specified in 5.1, shall be packed in a style RSC, fiberboard box conforming to V3c of PPP-B-636. The box shall be closed and sealed in accordance with the appendix thereto.

5.2.2 Packing, level C. Filter elements, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Shipping containers shall be in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

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5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with Drawing D150-2-1. All markings on the shipping containers shall be clear, legible and permanent after the application of the wood preservative specified in 5.2.

6. NOTES

6.1 Intended use. The filter elements covered by this specification are intended for use as a component of the M11 and M14 protective mask canisters.

6.2 Ordering data. Procurememt documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of filter material to be used.
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Supplementary manufacturing information. The following additional manufacturing information has been found helpful in producing a satisfactory filter element. This information is furnished with the distinct understanding:

(a) That the supplier is not required to use this information.
(b) That the provision of the specification must be met regardless of the method of manufacture.
(c) That the Government makes no warranty that material made using this information will pass the specification requirements.

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Adhesive. An adhesive suitable for cementing the filter has been prepared according to the following formulation:

Ingredients	Parts by weight
Latex, 60 percent centrifuged (Firestone Rubber & Latex Co.)	810
Ten percent solution NaOH	20
Ten percent solution Alkanol (Dupont)	25
Forty percent solution formaldehyde	20
*Vulcanizing Dispersion C95 (R. T. Vanderbilt Co., Inc.)	50

*Added just prior to use of cement.

Creasing. The filter material should be creased just deep enough to serve as a guide for the subsequent folding operation. Deep creasing will weaken or tear the filter material.

Number of pleats. When relatively thick filter material is used in production of the filter element, it is generally advantageous to use a lower number of pleats in order to meet the air resistance requirement. When relatively thin material is used, it is generally advantageous to use a higher number of pleats in order to meet the aerosol penetration requirement.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 4240-A414

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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